DO NOW · REQUIRED PRACTICAL

Density of a regular solid

Particle model of matter
Paper 1 · Combined and Separate
4 steps

A student measures the density of a rectangular metal block.

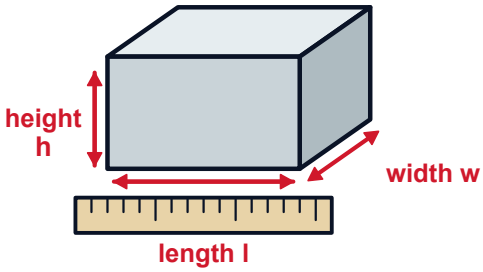
1 **Measure the mass of the block with a balance.**

metal block
mass m = ___ g



measure the mass

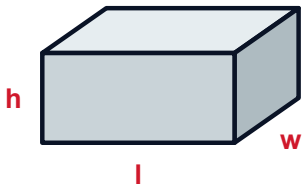
2 **Measure the length, width and height of the block with a ruler.**



3 **Calculate the volume: length × width × height.**

$$V = l \times w \times h$$

volume (cm³) = length × width × height



4 **Calculate the density: mass ÷ volume.**

$$\rho = m \div V$$

density (g/cm³) = mass (g) ÷ volume (cm³)

use m from step 1 and V from step 3

Note: The two measuring steps can be done in either order.

© 2026 PanPhy Labs. All rights reserved.